

3 - CI/CD Gitlab

Name of report: `CICD_GITLAB_LAB_3_Nikita_Niakhai`

Course: DevOps

Performed by *Nikita Niakhai* student number 19

Date submission: 12.02.2026

Task 1: Infra Deployment

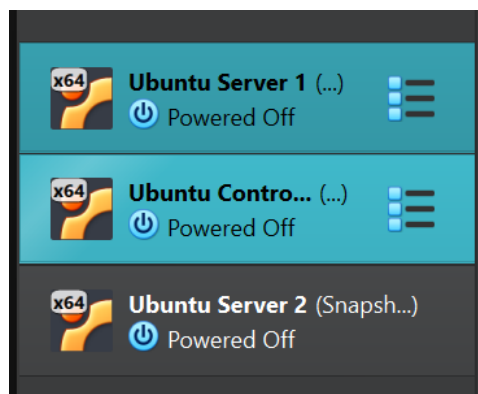
1. Deployed three VMs that I will be using as Gitlab Server, Gitlab Runner, and the deployment server.

My domain is `st19.sne.com`

`192.168.30.101` — ubuntu desktop with gitlab and postfix and domain `VM1`

`192.168.30.102` — ubuntu server 2 with Gitlab Runner `VM2`

`192.168.30.103` — ubuntu server 1 with Deployment Server `VM3`



Hosts configuration is done on all 3 VMs, here is example fot VM1:

```
sudo nano /etc/hosts
```

```
^^^
```

Add these lines:

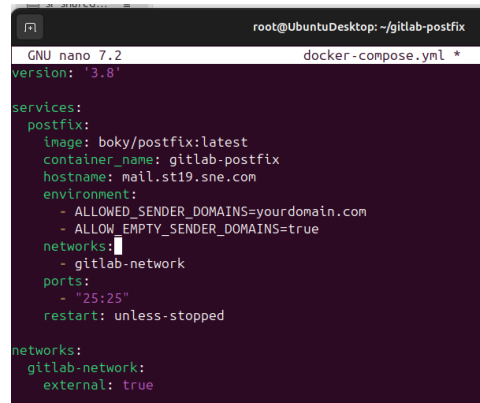
```
^^^
```

```
192.168.30.101    st19.sne.com gitlab
192.168.30.102    runner.st19.sne.com runner
192.168.30.103    deploy.st19.sne.com deploy
```

Installed docker network to communicate on **VM1** :

```
docker network create gitlab-network
```

Installed postfix for docker so that it will be communicating with gitlab runner on another container.



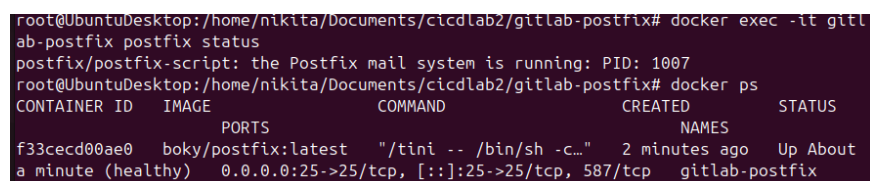
```
root@UbuntuDesktop: ~/gitlab-postfix
GNU nano 7.2 docker-compose.yml *
version: '3.8'

services:
  postfix:
    image: boky/postfix:latest
    container_name: gitlab-postfix
    hostname: mail.st19.sne.com
    environment:
      - ALLOWED_SENDER_DOMAINS=yourdomain.com
      - ALLOW_EMPTY_SENDER_DOMAINS=true
    networks:
      - gitlab-network
    ports:
      - "25:25"
    restart: unless-stopped

networks:
  gitlab-network:
    external: true
```

Docker compose file for postfix container

Then I executed docker image with postfix and here are result of postfix running



```
root@UbuntuDesktop: /home/nikita/Documents/cicdlab2/gitlab-postfix# docker exec -it gitlab-postfix postfix status
postfix/postfix-script: the Postfix mail system is running: PID: 1007
root@UbuntuDesktop: /home/nikita/Documents/cicdlab2/gitlab-postfix# docker ps
CONTAINER ID   IMAGE               COMMAND                  CREATED        STATUS
f33cecd00ae0   boky/postfix:latest "/tini -- /bin/sh -c..." 2 minutes ago  Up About
a minute (healthy)   0.0.0.0:25->25/tcp, [::]:25->25/tcp, 587/tcp   gitlab-postfix
```

2. Set up Gitlab Server (VM1)

For my gitlab container I added these envs:

```
services:
  gitlab:
    image: gitlab/gitlab-ce:latest
    container_name: gitlab
    hostname: st19.sne.com
    environment:
      GITLAB_OMNIBUS_CONFIG: |
        external_url 'http://st19.sne.com'
        gitlab_rails['smtp_enable'] = true
        gitlab_rails['smtp_address'] = "gitlab-postfix"
        gitlab_rails['smtp_port'] = 25
        gitlab_rails['smtp_domain'] = "yourdomain.com"
        gitlab_rails['smtp_tls'] = false
        gitlab_rails['smtp_openssl_verify_mode'] = 'none'
        gitlab_rails['smtp_enable_starttls_auto'] = false
        gitlab_rails['smtp_authentication'] = false
        gitlab_rails['gitlab_email_from'] = 'gitlab@yourdomain.com'
        gitlab_rails['gitlab_email_reply_to'] = 'noreply@yourdomain.com'
    networks:
      - gitlab-network
    ports:
      - "80:80"
      - "443:443"
      - "22:22"
    volumes:
      - gitlab-config:/etc/gitlab
      - gitlab-logs:/var/log/gitlab
```

```
- gitlab-data:/var/opt/gitlab
restart: unless-stopped
```

```
networks:
  gitlab-network:
    external: true
```

```
volumes:
  gitlab-config:
  gitlab-logs:
  gitlab-data:
```

Ports 22 80 443 were unused, I checked that with `netstat -tlnp`

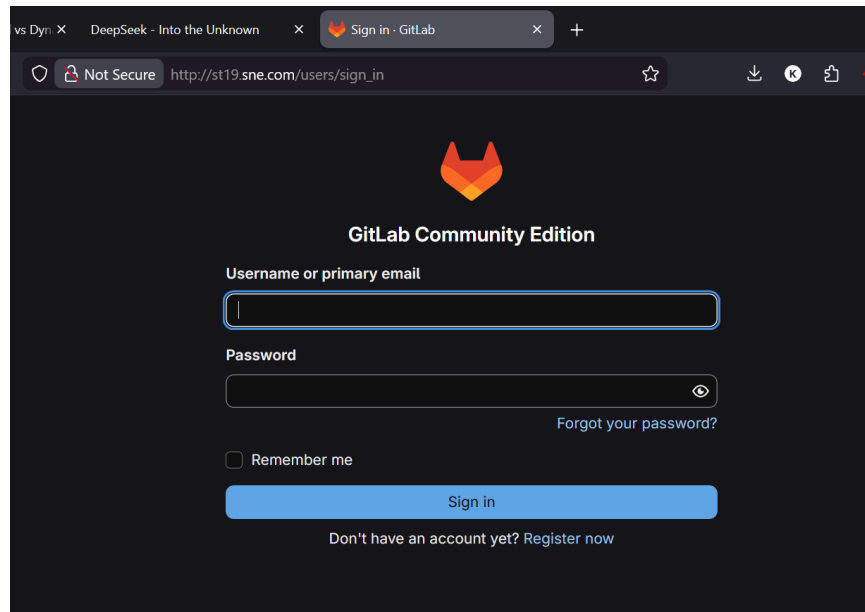
```
root@UbuntuDesktop:/home/nikita/Documents/cicdlab2# netstat -tlnp
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
gram name
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN
psd
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN
tend-resolve
tcp        0      0 127.0.0.54:53           0.0.0.0:*               LISTEN
tend-resolve
tcp        0      0 0.0.0.0:25              0.0.0.0:*               LISTEN
cker-proxy
tcp6       0      0 :::631                  :::*                     LISTEN
psd
tcp6       0      0 :::25                   :::*                     LISTEN
cker-proxy
```

Then I runed my docker container and checked whether 2 container are in the same docker network using `docker network inspect gitlab-network`

```
root@UbuntuDesktop:/home/nikita/Documents/cicdlab2/gitlab# docker ps
CONTAINER ID   IMAGE                                PORTS          COMMAND                  CREATED    STAT
US            gitlab/gitlab-ce:latest             /assets/init-conta...  5 minutes ago    Up 5
minutes (health: starting)  0.0.0.0:22->22/tcp, [::]:22->22/tcp, 0.0.0.0:80->80/tcp,
[::]:80->80/tcp, 0.0.0.0:443->443/tcp, [::]:443->443/tcp  st19-gitlab
f33cecd00ae0   bokyo/postfix:latest                "/tini -- /bin/sh -c..." 36 minutes ago    Up 3
6 minutes (healthy)        0.0.0.0:25->25/tcp, [::]:25->25/tcp, 587/tcp
gitlab-postfix

root@UbuntuDesktop:/home/nikita/Documents/cicdlab2/gitlab# docker network inspect gitlab-network
[{"Name": "gitlab-network", "Labels": {}, "Options": {}, "Containers": {"879fcac8030773ccbc989347e9af44495a0513e3ddf48ff27e57045f9d7460b4": {"Name": "st19-gitlab", "EndpointID": "4499de1caa0719a18705dd55cc0bab559d5721537568d656811661de", "MacAddress": "3e:04:20:c2:4b:d7", "IPv4Address": "172.18.0.3/16", "IPv6Address": ""}, "f33cecd00ae0b12b962b9378ba33111dafdad48420a62620ba3c84a020b450a7": {"Name": "gitlab-postfix", "EndpointID": "448b49148da4d0aecbdca12481dca4a52a0a2cd84854238864db4b3b", "MacAddress": "8e:30:58:73:a8:1a", "IPv4Address": "172.18.0.2/16", "IPv6Address": ""}}}]
```

Now I can sign in via my host machine:



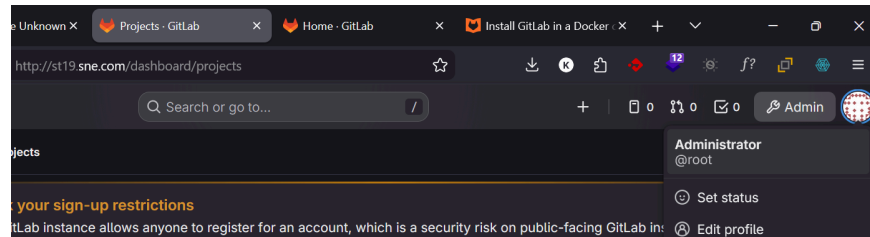
I found out what means full local gitlab solution: even accounts and passwords are all local :) to understand that I used AI and checked volumes created and mounted, because I thought maybe the error with them.

```
root@UbuntuDesktop: /home/nikita/Documents/cicdlab2/gitlab# docker volume ls
DRIVER      VOLUME NAME
local       7a127189add7faa9f5847edd52840f9291036bc2d170bd875e846c125cd68a3c
local       94c6ae88a87e40c218dda9d7f2887356db654bde95f70811c81a178ad3b1bf1b
local       36370b12f567ff789242d75f2f22e130778f1357f8e87ef148896cbecd846bac
local       gitlab_gitlab-config
local       gitlab_gitlab-data
local       gitlab_gitlab-logs
root@UbuntuDesktop: /home/nikita/Documents/cicdlab2/gitlab#
```

To sign in as root and find credentials I used docker docs info [x]. I visited the GitLab URL, and sign in with the username `root` and the password from the following command:

```
sudo docker exec -it gitlab grep 'Password:' /etc/gitlab/initial_root_password
```

```
gitlab(p190)> exit
root@UbuntuDesktop:/home/nikita/Documents/cicdlab2/gitlab# docker exec -it st19-
gitlab grep 'Password:' /etc/gitlab/initial_root_password
Password: CtqoL7UE04MNAiwVfQb0Ay2N3lpmoHVV1Jd4xU43Pn0=
root@UbuntuDesktop:/home/nikita/Documents/cicdlab2/gitlab#
```



Then I did some default profile configurations.

Added SSH key generation:

```
# Generate SSH key pair
ssh-keygen -t ed25519 -C "st19@sne.com" -f ~/.ssh/gitlab_st19
```

```
# Paste your public key in the Key field
cat ~/.ssh/gitlab_st19.pub
```

```
# Create/edit SSH config
nano ~/.ssh/config
'''
```

```
Add this configuration:
'''
```

```
Host st19.sne.com
  HostName st19.sne.com
  User git
  IdentityFile ~/.ssh/gitlab_st19
  StrictHostKeyChecking no
```

```
chmod 700 ~/.ssh
chmod 600 ~/.ssh/gitlab_st19
```

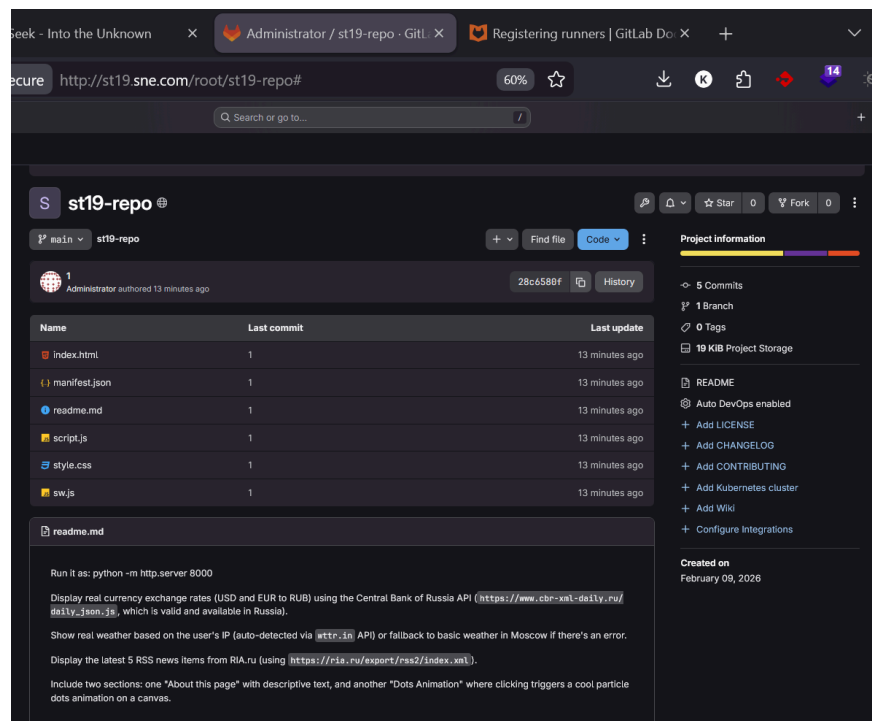
```
chmod 644 ~/.ssh/gitlab_st19.pub
chmod 600 ~/.ssh/config
```

```
ssh -T git@st19.sne.com
```

```
root@UbuntuDesktop:~# ssh -T git@st19.sne.com
Enter passphrase for key '/root/.ssh/gitlab_st19':
Welcome to GitLab, @root!
root@UbuntuDesktop:~#
```

After all prerequisites on host machines and gitlab server were arranged I pushed my project from local env to the lab:

```
# Or if already cloned via HTTP, change remote:
cd st19-repo
git remote set-url origin git@st19.sne.com:root/st19-repo.git
```



Get registration token from GitLab

[st19.sne.com/root/st19-repo/-/settings/ci_cd#js-runners-settings]

GR1348941-k3eXvejPHSJdNUQqiGz

3. Gitlab runner on VM2 setup on clean OS.

Installed docker runner [x]

```
curl -L "https://packages.gitlab.com/install/repositories/runner/gitlab-runner/script.deb.sh" -o script.deb.sh
```

```
less script.deb.sh
```

```
sudo bash script.deb.sh
```

```
sudo apt install gitlab-runner
```

```
# registering runner
```

```
sudo gitlab-runner register
```

```
# Runner tag to st19-runner
```

```
# set the executor type to shell
```

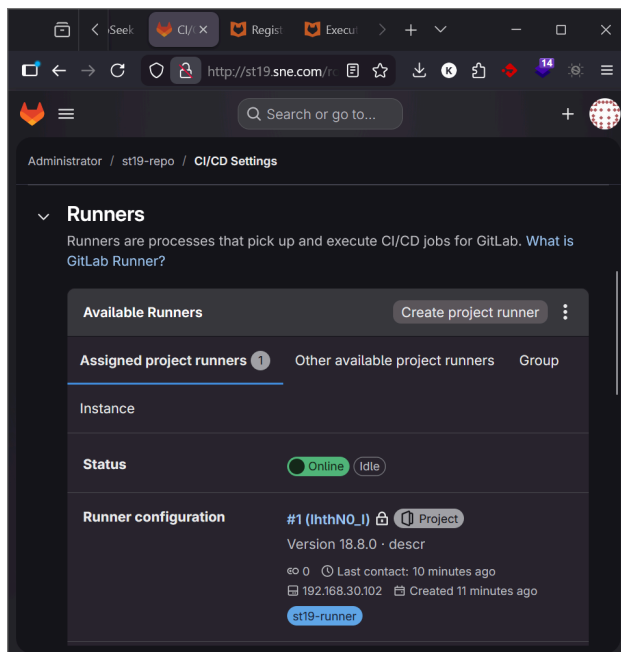
I authenticate Gitlab runner with Gitlab server via registration token, and validated:


```

root@UbuntuServer2:~# sudo gitlab-runner register
Runtime platform arch=amd64 os=linux pid=10961 revision=9ffb4aa0 version=18.8.0
Running in system-mode.

Enter the GitLab instance URL (for example, https://gitlab.com/):
http://st19.sne.com
Enter the registration token:
GR13699411-k3eKveJP4SJDNU0q1gZ
Enter a description for the runner:
[UbuntuServer2]: descr
Enter tags for the runner (comma-separated):
st19-runner
Enter optional maintenance note for the runner:
note
WARNING: Support for registration tokens and runner parameters in the 'register' command has been deprecated in GitLab
port for authentication tokens. For more information, see https://docs.gitlab.com/ci/runners/new_creation_workflow/
Registering runner... succeeded correlation_id=01KH1TSF0MR51MM4FV9K7YD2AH runners=k3eKveJP
Enter an executor: kubernetes, docker-autoscaler, instance, custom, virtualbox, docker, docker+machine, shell, ssh, p
shell
Runner registered successfully. Feel free to start it, but if it's running already the config should be automatically
Configuration (with the authentication token) was saved in "/etc/gitlab-runner/config.toml"
root@UbuntuServer2:~#

```



```

root@UbuntuServer2:~# cat /etc/gitlab-runner/config.toml
concurrent = 1
check_interval = 0
connection_max_age = "15m0s"
shutdown_timeout = 0

[session_server]
session_timeout = 1800

[[runners]]
name = "descr"
url = "http://st19.sne.com"
id = 1
token = "g1rtr-1hthN0_1RwTg8vqaErIdw86Mqpw0JEkd0ozDw.01.121ct5rdd"
token_obtained_at = 2026-02-09T18:42:20Z
token_expires_at = 0001-01-01T00:00:00Z
executor = "shell"

[runners.cache]
MaxUploadeArchIvsSize = 0
[runners.cache.s3]
[runners.cache.gcs]
[runners.cache.azure]
root@UbuntuServer2:~#

```

GitLab executor is an instance that actually runs my CI/CD pipeline, in this case it is set on **VM2**, so all commands for pipeline will be executed on this machines, also since we use **shell** as executor. If we'd use **container** - for each jib it would create clean isolated container; **ssh** - would execute commands on different machine and etc.

4. Set up the Deployment Server (VM3).

Here I need to set up **SSH key-based authentication** so that **VM2** (GitLab Runner) can connect to **VM3** (Deployment Server) without a password to deploy

applications:

VM3

```
# Make sure your user can run docker
sudo usermod -aG default_user_not_root $USER
```

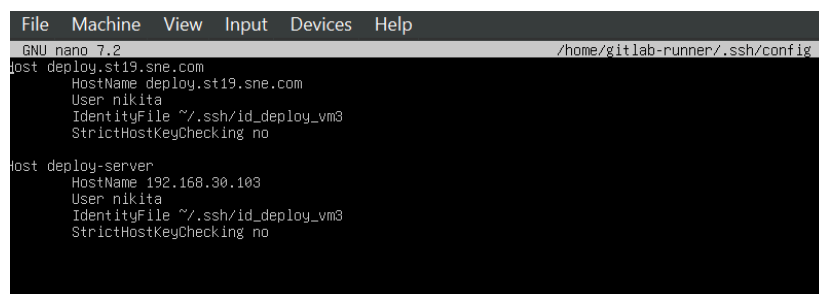
VM2

```
# Switch to gitlab-runner user
sudo su - gitlab-runner

# Generate SSH key (no passphrase for automation)
ssh-keygen -t ed25519 -C "gitlab-runner-deploy" -f ~/.ssh/
id_deploy_vm3

# ssh-copy-id to vm3 my key id_deploy_vm3

# added hosts configuration to seamless deploy connection
```



```
File  Machine  View  Input  Devices  Help
GNU nano 7.2 /home/gitlab-runner/.ssh/config
Host deploy.st19.sne.com
  HostName deploy.st19.sne.com
  User nikita
  IdentityFile ~/.ssh/id_deploy_vm3
  StrictHostKeyChecking no

Host deploy-server
  HostName 192.168.30.103
  User nikita
  IdentityFile ~/.ssh/id_deploy_vm3
  StrictHostKeyChecking no
```

`ssh deploy-server` to connect from runner to deploy env

Task 2: Create CI/CD Pipeline

1. Created single page application and pushed it to project:

```
root@UbuntuDesktop: /home/nikita/Documents/cicdlab2# tree
.
├── gitlab
│   └── docker-compose.yml
├── gitlab-postfix
│   └── docker-compose.yml
├── pass_root.pem
├── simple_pwa
│   ├── ansible
│   │   ├── deploy-playbook.yml
│   │   └── inventory.ini
│   ├── copy_gitlab.yml
│   ├── Dockerfile
│   ├── index.html
│   ├── manifest.json
│   ├── readme.md
│   ├── script.js
│   ├── style.css
│   └── sw.js
```

2. GitLab project:

The screenshot shows the GitLab interface for a project named 'st19-repo'. At the top, there's a header with the project name, a search bar, and buttons for 'Find file', 'Code', and 'History'. Below the header, a commit message 'fix ansible ssh error' is displayed, along with a commit hash '8b97cf69' and a 'History' button. A table lists the project's files and their commit history:

Name	Last commit	Last update
ansible	fix ansible ssh error	1 day ago
gitlab-ci.yml	add ci cd files	2 days ago
gitlab-ci.yml	Update .gitlab-ci.yml file	1 day ago
Dockerfile	add ci cd files	2 days ago
copy_gitlab.yml	ansible added	1 day ago
index.html	1	2 days ago
manifest.json	1	2 days ago
readme.md	1	2 days ago
script.js	1	2 days ago
style.css	1	2 days ago
sw.js	1	2 days ago

Below the table, the 'readme.md' file content is visible, detailing the application's functionality: displaying real currency exchange rates, showing real weather based on the user's IP, displaying the latest 5 RSS news items from RIA.ru, and including two sections: 'About this page' and 'Dots Animation'.

On the right side, 'Project information' is shown, including 16 commits, 3 branches, 0 tags, 346 KiB project storage, and 1 environment. It also lists actions like 'Add LICENSE', 'Add CHANGELOG', 'Add CONTRIBUTING', 'Auto DevOps enabled', 'Add Kubernetes cluster', 'Add Wiki', and 'Configure Integrations'. The project was created on February 09, 2026.

3. Creating CI/CD.

I registered on docker hub. Created variables inside project for pushing to docker-hub:

CI/CD Variables </> 2	
Key ↑	Value
DOCKER_HUB_PASSWORD  Protected Masked	 
DOCKER_HUB_USERNAME  Protected Masked	 

I created `Dockerfile`:

```
FROM nginx:alpine

# Copy SPA files to nginx html directory
COPY . /usr/share/nginx/html/

# Expose port 80
EXPOSE 80

# Start nginx
CMD ["nginx", "-g", "daemon off;"]
```

Then I created my `.gitlab-ci.yml`:

```
stages:
  - build
  - test
  - docker-build
  - docker-push
  - deploy
```

```

variables:
  DOCKER_IMAGE_NAME: "nikitanekhay/st19-spa"
  DOCKER_IMAGE_TAG: "${CI_COMMIT_SHORT_SHA}"

# CI Stage 1: Build the application
build-app:
  stage: build
  tags:
    - st19-runner
  script:
    - echo "Building the application..."
    - ls -la
    # If using npm:
    # - npm install
    # - npm run build
    - echo "Build completed successfully"
  artifacts:
    paths:
      - . # Save all files for next stages
    expire_in: 1 hour

# CI Stage 2: Run tests
test-app:
  stage: test
  tags:
    - st19-runner
  dependencies:
    - build-app
  script:
    - echo "Running application tests..."
    # Add your actual test commands here
    # For npm projects:
    # - npm test
    # For simple validation:
    - test -f index.html && echo "index.html exists - OK"
    - echo "All tests passed!"

```

```

# CI Stage 3: Build Docker image
build-docker:
  stage: docker-build
  tags:
    - st19-runner
  dependencies:
    - build-app
  script:
    - echo "Building Docker image..."
    - docker build -t ${DOCKER_IMAGE_NAME}:${DOCKER_IMAGE_TAG} .
    - docker build -t ${DOCKER_IMAGE_NAME}:latest .
    - echo "Docker image built successfully"
    - docker images | grep st19-spa

# CI Stage 4: Push to Docker Hub
push-docker:
  stage: docker-push
  tags:
    - st19-runner
  dependencies:
    - build-docker
  script:
    - echo "Logging into Docker Hub..."
    - echo "$DOCKER_HUB_PASSWORD" | docker login -u "$DOCKER_HUB_USERNAME" --password-stdin
    - echo "Pushing Docker image to Docker Hub..."
    - docker push ${DOCKER_IMAGE_NAME}:${DOCKER_IMAGE_TAG}
    - docker push ${DOCKER_IMAGE_NAME}:latest
    - echo "Docker image pushed successfully"
    - docker logout

# CD Stage 1: Deploy to deployment server
deploy-app:
  stage: deploy

```

```

tags:
  - st19-runner
dependencies: []
script:
  - echo "Deploying to VM3 (Deployment Server)..."

  # Stop and remove old container if exists
  - ssh deploy-server "docker stop st19-spa || true"
  - ssh deploy-server "docker rm st19-spa || true"

  # Pull latest image
  - echo "Pulling Docker image on deployment server..."
  - ssh deploy-server "docker pull ${DOCKER_IMAGE_NAME}:latest"

  # Run new container
  - echo "Starting new container..."
  - ssh deploy-server "docker run -d --name st19-spa -p 8080:80 ${DOCKER_IMAGE_NAME}:latest"

  # Verify deployment
  - echo "Verifying deployment..."
  - ssh deploy-server "docker ps | grep st19-spa"
  - echo "Deployment completed successfully!"
  - echo "Access the application at: http://deploy.st19.sne.com:8080"
only:
  - main

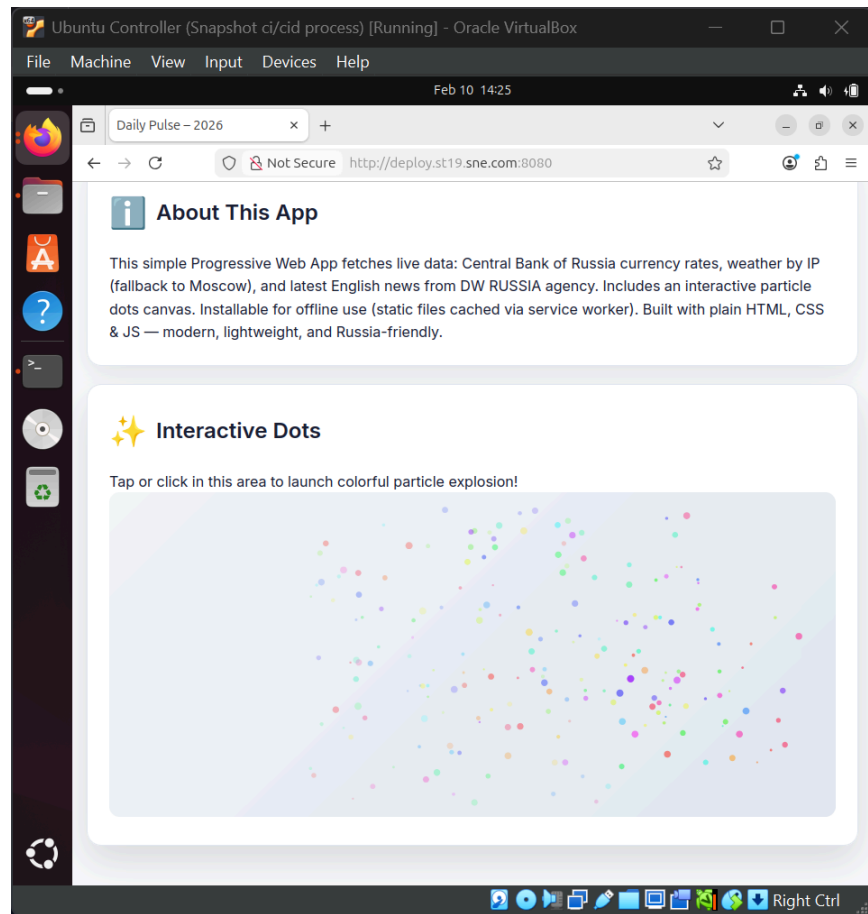
```

Here I got error on CI/CD, because docker was not installed on `VM3`, so I installed.

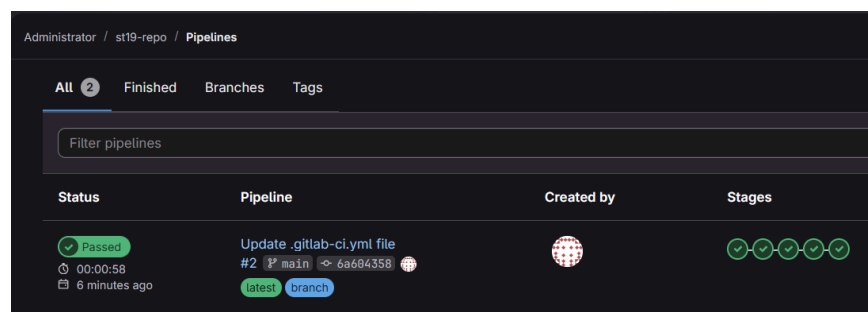
4. I validate that the deployment is successful by accessing the web app via the browser on deployment server

side.

Here Is my project:

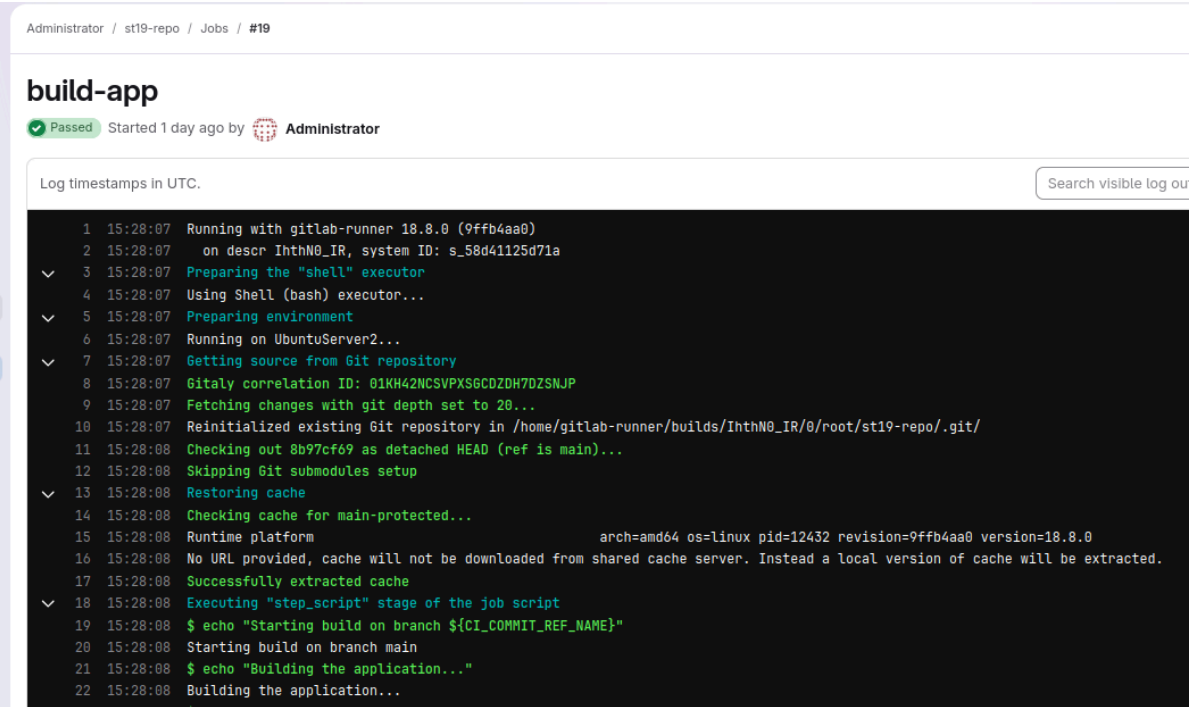


This is successful results for CI/CD pipeline:



Here is more detailed description in a screenshot for each stage:

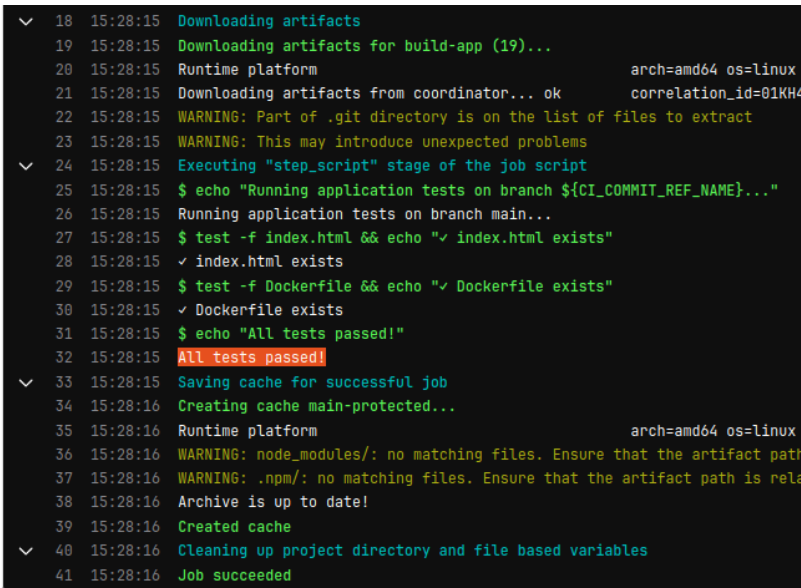
- Build the application



The screenshot shows the GitLab CI/CD interface for a job named 'build-app'. The job status is 'Passed' and it was started 1 day ago by 'Administrator'. The log shows the following steps:

```
1 15:28:07 Running with gitlab-runner 18.8.0 (9ffb4aa0)
2 15:28:07 on descr InthN0_IR, system ID: s_58d41125d71a
3 15:28:07 Preparing the "shell" executor
4 15:28:07 Using Shell (bash) executor...
5 15:28:07 Preparing environment
6 15:28:07 Running on UbuntuServer2...
7 15:28:07 Getting source from Git repository
8 15:28:07 Gitaly correlation ID: 01KH42NCSVPXS6CDZDH7DZSNJP
9 15:28:07 Fetching changes with git depth set to 20...
10 15:28:07 Reinitialized existing Git repository in /home/gitlab-runner/builds/InthN0_IR/0/root/st19-repo/.git/
11 15:28:08 Checking out 8b97cf69 as detached HEAD (ref is main)...
12 15:28:08 Skipping Git submodules setup
13 15:28:08 Restoring cache
14 15:28:08 Checking cache for main-protected...
15 15:28:08 Runtime platform arch=amd64 os=linux pid=12432 revision=9ffb4aa0 version=18.8.0
16 15:28:08 No URL provided, cache will not be downloaded from shared cache server. Instead a local version of cache will be extracted.
17 15:28:08 Successfully extracted cache
18 15:28:08 Executing "step_script" stage of the job script
19 15:28:08 $ echo "Starting build on branch ${CI_COMMIT_REF_NAME}"
20 15:28:08 Starting build on branch main
21 15:28:08 $ echo "Building the application..."
22 15:28:08 Building the application...
```

- Run test (to check the application works ok)



The screenshot shows the GitLab CI/CD interface for a job named 'test'. The log shows the following steps:

```
18 15:28:15 Downloading artifacts
19 15:28:15 Downloading artifacts for build-app (19)...
20 15:28:15 Runtime platform arch=amd64 os=linux
21 15:28:15 Downloading artifacts from coordinator... ok correlation_id=01KH4
22 15:28:15 WARNING: Part of .git directory is on the list of files to extract
23 15:28:15 WARNING: This may introduce unexpected problems
24 15:28:15 Executing "step_script" stage of the job script
25 15:28:15 $ echo "Running application tests on branch ${CI_COMMIT_REF_NAME}..."
26 15:28:15 Running application tests on branch main...
27 15:28:15 $ test -f index.html && echo "✓ index.html exists"
28 15:28:15 ✓ index.html exists
29 15:28:15 $ test -f Dockerfile && echo "✓ Dockerfile exists"
30 15:28:15 ✓ Dockerfile exists
31 15:28:15 $ echo "All tests passed!"
32 15:28:15 All tests passed!
33 15:28:15 Saving cache for successful job
34 15:28:16 Creating cache main-protected...
35 15:28:16 Runtime platform arch=amd64 os=linux
36 15:28:16 WARNING: node_modules/: no matching files. Ensure that the artifact path is rel
37 15:28:16 WARNING: .npm/: no matching files. Ensure that the artifact path is rel
38 15:28:16 Archive is up to date!
39 15:28:16 Created cache
40 15:28:16 Cleaning up project directory and file based variables
41 15:28:16 Job succeeded
```

- Build docker image (Note: you need Dockerfile)

```

24 15:28:18 Executing "step_script" stage of the job script
25 15:28:18 $ echo "Building Docker image for branch ${CI_COMMIT_REF_NAME}..."
26 15:28:18 Building Docker image for branch main...
27 15:28:18 $ if [ "${CI_COMMIT_REF_NAME}" = "main" ]; then # collapsed multi-line command
28 15:28:19 #0 building with "default" instance using docker driver
29 15:28:19 #1 [internal] load build definition from Dockerfile
30 15:28:19 #1 transferring dockerfile: 132B done
31 15:28:19 #1 DONE 0.0s
32 15:28:19 #2 [internal] load metadata for docker.io/library/nginx:alpine
33 15:28:21 #2 DONE 2.2s
34 15:28:21 #3 [internal] load .dockerignore
35 15:28:21 #3 transferring context: 2B done
36 15:28:21 #3 DONE 0.0s
37 15:28:21 #4 [internal] load build context
38 15:28:21 #4 transferring context: 35.66kB 0.0s done
39 15:28:21 #4 DONE 0.1s
40 15:28:21 #5 [1/2] FROM docker.io/library/nginx:alpine@sha256:1d13701a5f9f3fb01aaa88cef234
41 15:28:21 #5 resolve docker.io/library/nginx:alpine@sha256:1d13701a5f9f3fb01aaa88cef234d6
42 15:28:21 #5 DONE 0.1s
43 15:28:21 #5 [1/2] FROM docker.io/library/nginx:alpine@sha256:1d13701a5f9f3fb01aaa88cef234
44 15:28:21 #5 CACHED
45 15:28:21 #6 [2/2] COPY . /usr/share/nginx/html
46 15:28:21 #6 DONE 0.3s
47 15:28:22 #7 exporting to image
48 15:28:22 #7 exporting layers
49 15:28:22 #7 exporting layers 0.3s done
50 15:28:22 #7 exporting manifest sha256:3a58487619a1ccc9748ddaa7013ea49fee3abc8f8150dd2f2166
51 15:28:22 #7 exporting manifest sha256:3a58487619a1ccc9748ddaa7013ea49fee3abc8f8150dd2f2166
52 15:28:22 #7 exporting config sha256:d8b9acff99cf0e27b4133f5856b0ca44b08dee79dfa051efdad3
53 15:28:22 #7 exporting attestation manifest sha256:2c2972e422f1af9786de18d247bf5a6b64f907c
54 15:28:22 #7 exporting manifest list sha256:f426474cc53f04d5d10035308e86cc61f9c0084353f4d7
55 15:28:22 #7 exporting manifest list sha256:f426474cc53f04d5d10035308e86cc61f9c0084353f4d7
56 15:28:22 #7 naming to docker.io/[MASKED]/st19-spa:8b97cf69 0.0s done

```

- Pushed to my docker hub account

```

5 15:28:30 Preparing environment
6 15:28:30 Running on UbuntuServer2...
7 15:28:31 Getting source from Git repository
8 15:28:31 Gitly correlation ID: 01KH42PSR9SGAV9WQCKXS4PM3V
9 15:28:31 Fetching changes with git depth set to 20...
10 15:28:31 Reinitialized existing Git repository in /home/gitlab-runner/builds/IntHNO_IR/0/root/st19-spa-repo/
11 15:28:31 Checking out 8b97cf69 as detached HEAD (ref is main)...
12 15:28:31 Skipping Git submodules setup
13 15:28:31 Restoring cache
14 15:28:31 Checking cache for main-protected...
15 15:28:31 Runtime platform arch=amd64 os=linux pid=12995 revision=9ffbb
16 15:28:31 No URL provided, cache will not be downloaded from shared cache server. Instead a local version
17 15:28:31 Successfully extracted cache
18 15:28:31 Executing "step_script" stage of the job script
19 15:28:31 $ echo "Logging into Docker Hub..."
20 15:28:31 Logging into Docker Hub...
21 15:28:31 $ echo "$DOCKER_HUB_PASSWORD" | docker login -u "$DOCKER_HUB_USERNAME" --password-stdin
22 15:28:33 WARNING! Your credentials are stored unencrypted in /home/gitlab-runner/.docker/config.json.
23 15:28:33 Configure a credential helper to remove this warning. See
24 15:28:33 https://docs.docker.com/go/credential-store/
25 15:28:33 Login Succeeded
26 15:28:33 $ echo "Pushing Docker image for branch ${CI_COMMIT_REF_NAME}..."
27 15:28:33 Pushing Docker image for branch main...
28 15:28:33 $ if [ "${CI_COMMIT_REF_NAME}" = "main" ]; then # collapsed multi-line command
29 15:28:34 The push refers to repository [docker.io/[MASKED]/st19-spa]
30 15:28:34
31 15:28:34
32 15:28:34
33 15:28:34
34 15:28:34
35 15:28:34
36 15:28:34
37 15:28:34
38 15:28:34
39 15:28:34
40 15:28:34
41 15:28:34
42 15:28:34
43 15:28:34
44 15:28:34
45 15:28:34
46 15:28:34
47 15:28:34
48 15:28:34
49 15:28:34
50 15:28:34
51 15:28:34
52 15:28:34
53 15:28:34
54 15:28:34
55 15:28:34
56 15:28:34
57 15:28:34
58 15:28:34
59 15:28:34
60 15:28:34
61 15:28:34
62 15:28:34
63 15:28:34
64 15:28:34
65 15:28:34
66 15:28:34
67 15:28:34
68 15:28:34
69 15:28:34
70 15:28:34
71 15:28:34
72 15:28:34
73 15:28:34
74 15:28:34
75 15:28:34
76 15:28:34
77 15:28:34
78 15:28:34
79 15:28:34
80 15:28:34
81 15:28:34
82 15:28:34
83 15:28:34
84 15:28:34
85 15:28:34
86 15:28:34
87 15:28:34
88 15:28:34
89 15:28:34
90 15:28:34
91 15:28:34
92 15:28:34
93 15:28:34
94 15:28:34
95 15:28:34
96 15:28:34
97 15:28:34
98 15:28:34
99 15:28:34
100 15:28:34
101 15:28:34
102 15:28:34
103 15:28:34
104 15:28:34
105 15:28:34
106 15:28:34
107 15:28:34
108 15:28:34
109 15:28:34
110 15:28:34
111 15:28:34
112 15:28:34
113 15:28:34
114 15:28:34
115 15:28:34
116 15:28:34
117 15:28:34
118 15:28:34
119 15:28:34
120 15:28:34
121 15:28:34
122 15:28:34
123 15:28:34
124 15:28:34
125 15:28:34
126 15:28:34
127 15:28:34
128 15:28:34
129 15:28:34
130 15:28:34
131 15:28:34
132 15:28:34
133 15:28:34
134 15:28:34
135 15:28:34
136 15:28:34
137 15:28:34
138 15:28:34
139 15:28:34
140 15:28:34
141 15:28:34
142 15:28:34
143 15:28:34
144 15:28:34
145 15:28:34
146 15:28:34
147 15:28:34
148 15:28:34
149 15:28:34
150 15:28:34
151 15:28:34
152 15:28:34
153 15:28:34
154 15:28:34
155 15:28:34
156 15:28:34
157 15:28:34
158 15:28:34
159 15:28:34
160 15:28:34
161 15:28:34
162 15:28:34
163 15:28:34
164 15:28:34
165 15:28:34
166 15:28:34
167 15:28:34
168 15:28:34
169 15:28:34
170 15:28:34
171 15:28:34
172 15:28:34
173 15:28:34
174 15:28:34
175 15:28:34
176 15:28:34
177 15:28:34
178 15:28:34
179 15:28:34
180 15:28:34
181 15:28:34
182 15:28:34
183 15:28:34
184 15:28:34
185 15:28:34
186 15:28:34
187 15:28:34
188 15:28:34
189 15:28:34
190 15:28:34
191 15:28:34
192 15:28:34
193 15:28:34
194 15:28:34
195 15:28:34
196 15:28:34
197 15:28:34
198 15:28:34
199 15:28:34
200 15:28:34
201 15:28:34
202 15:28:34
203 15:28:34
204 15:28:34
205 15:28:34
206 15:28:34
207 15:28:34
208 15:28:34
209 15:28:34
210 15:28:34
211 15:28:34
212 15:28:34
213 15:28:34
214 15:28:34
215 15:28:34
216 15:28:34
217 15:28:34
218 15:28:34
219 15:28:34
220 15:28:34
221 15:28:34
222 15:28:34
223 15:28:34
224 15:28:34
225 15:28:34
226 15:28:34
227 15:28:34
228 15:28:34
229 15:28:34
230 15:28:34
231 15:28:34
232 15:28:34
233 15:28:34
234 15:28:34
235 15:28:34
236 15:28:34
237 15:28:34
238 15:28:34
239 15:28:34
240 15:28:34
241 15:28:34
242 15:28:34
243 15:28:34
244 15:28:34
245 15:28:34
246 15:28:34
247 15:28:34
248 15:28:34
249 15:28:34
250 15:28:34
251 15:28:34
252 15:28:34
253 15:28:34
254 15:28:34
255 15:28:34
256 15:28:34
257 15:28:34
258 15:28:34
259 15:28:34
260 15:28:34
261 15:28:34
262 15:28:34
263 15:28:34
264 15:28:34
265 15:28:34
266 15:28:34
267 15:28:34
268 15:28:34
269 15:28:34
270 15:28:34
271 15:28:34
272 15:28:34
273 15:28:34
274 15:28:34
275 15:28:34
276 15:28:34
277 15:28:34
278 15:28:34
279 15:28:34
280 15:28:34
281 15:28:34
282 15:28:34
283 15:28:34
284 15:28:34
285 15:28:34
286 15:28:34
287 15:28:34
288 15:28:34
289 15:28:34
290 15:28:34
291 15:28:34
292 15:28:34
293 15:28:34
294 15:28:34
295 15:28:34
296 15:28:34
297 15:28:34
298 15:28:34
299 15:28:34
300 15:28:34
301 15:28:34
302 15:28:34
303 15:28:34
304 15:28:34
305 15:28:34
306 15:28:34
307 15:28:34
308 15:28:34
309 15:28:34
310 15:28:34
311 15:28:34
312 15:28:34
313 15:28:34
314 15:28:34
315 15:28:34
316 15:28:34
317 15:28:34
318 15:28:34
319 15:28:34
320 15:28:34
321 15:28:34
322 15:28:34
323 15:28:34
324 15:28:34
325 15:28:34
326 15:28:34
327 15:28:34
328 15:28:34
329 15:28:34
330 15:28:34
331 15:28:34
332 15:28:34
333 15:28:34
334 15:28:34
335 15:28:34
336 15:28:34
337 15:28:34
338 15:28:34
339 15:28:34
340 15:28:34
341 15:28:34
342 15:28:34
343 15:28:34
344 15:28:34
345 15:28:34
346 15:28:34
347 15:28:34
348 15:28:34
349 15:28:34
350 15:28:34
351 15:28:34
352 15:28:34
353 15:28:34
354 15:28:34
355 15:28:34
356 15:28:34
357 15:28:34
358 15:28:34
359 15:28:34
360 15:28:34
361 15:28:34
362 15:28:34
363 15:28:34
364 15:28:34
365 15:28:34
366 15:28:34
367 15:28:34
368 15:28:34
369 15:28:34
370 15:28:34
371 15:28:34
372 15:28:34
373 15:28:34
374 15:28:34
375 15:28:34
376 15:28:34
377 15:28:34
378 15:28:34
379 15:28:34
380 15:28:34
381 15:28:34
382 15:28:34
383 15:28:34
384 15:28:34
385 15:28:34
386 15:28:34
387 15:28:34
388 15:28:34
389 15:28:34
390 15:28:34
391 15:28:34
392 15:28:34
393 15:28:34
394 15:28:34
395 15:28:34
396 15:28:34
397 15:28:34
398 15:28:34
399 15:28:34
400 15:28:34
401 15:28:34
402 15:28:34
403 15:28:34
404 15:28:34
405 15:28:34
406 15:28:34
407 15:28:34
408 15:28:34
409 15:28:34
410 15:28:34
411 15:28:34
412 15:28:34
413 15:28:34
414 15:28:34
415 15:28:34
416 15:28:34
417 15:28:34
418 15:28:34
419 15:28:34
420 15:28:34
421 15:28:34
422 15:28:34
423 15:28:34
424 15:28:34
425 15:28:34
426 15:28:34
427 15:28:34
428 15:28:34
429 15:28:34
430 15:28:34
431 15:28:34
432 15:28:34
433 15:28:34
434 15:28:34
435 15:28:34
436 15:28:34
437 15:28:34
438 15:28:34
439 15:28:34
440 15:28:34
441 15:28:34
442 15:28:34
443 15:28:34
444 15:28:34
445 15:28:34
446 15:28:34
447 15:28:34
448 15:28:34
449 15:28:34
450 15:28:34
451 15:28:34
452 15:28:34
453 15:28:34
454 15:28:34
455 15:28:34
456 15:28:34
457 15:28:34
458 15:28:34
459 15:28:34
460 15:28:34
461 15:28:34
462 15:28:34
463 15:28:34
464 15:28:34
465 15:28:34
466 15:28:34
467 15:28:34
468 15:28:34
469 15:28:34
470 15:28:34
471 15:28:34
472 15:28:34
473 15:28:34
474 15:28:34
475 15:28:34
476 15:28:34
477 15:28:34
478 15:28:34
479 15:28:34
480 15:28:34
481 15:28:34
482 15:28:34
483 15:28:34
484 15:28:34
485 15:28:34
486 15:28:34
487 15:28:34
488 15:28:34
489 15:28:34
490 15:28:34
491 15:28:34
492 15:28:34
493 15:28:34
494 15:28:34
495 15:28:34
496 15:28:34
497 15:28:34
498 15:28:34
499 15:28:34
500 15:28:34
501 15:28:34
502 15:28:34
503 15:28:34
504 15:28:34
505 15:28:34
506 15:28:34
507 15:28:34
508 15:28:34
509 15:28:34
510 15:28:34
511 15:28:34
512 15:28:34
513 15:28:34
514 15:28:34
515 15:28:34
516 15:28:34
517 15:28:34
518 15:28:34
519 15:28:34
520 15:28:34
521 15:28:34
522 15:28:34
523 15:28:34
524 15:28:34
525 15:28:34
526 15:28:34
527 15:28:34
528 15:28:34
529 15:28:34
530 15:28:34
531 15:28:34
532 15:28:34
533 15:28:34
534 15:28:34
535 15:28:34
536 15:28:34
537 15:28:34
538 15:28:34
539 15:28:34
540 15:28:34
541 15:28:34
542 15:28:34
543 15:28:34
544 15:28:34
545 15:28:34
546 15:28:34
547 15:28:34
548 15:28:34
549 15:28:34
550 15:28:34
551 15:28:34
552 15:28:34
553 15:28:34
554 15:28:34
555 15:28:34
556 15:28:34
557 15:28:34
558 15:28:34
559 15:28:34
560 15:28:34
561 15:28:34
562 15:28:34
563 15:28:34
564 15:28:34
565 15:28:34
566 15:28:34
567 15:28:34
568 15:28:34
569 15:28:34
570 15:28:34
571 15:28:34
572 15:28:34
573 15:28:34
574 15:28:34
575 15:28:34
576 15:28:34
577 15:28:34
578 15:28:34
579 15:28:34
580 15:28:34
581 15:28:34
582 15:28:34
583 15:28:34
584 15:28:34
585 15:28:34
586 15:28:34
587 15:28:34
588 15:28:34
589 15:28:34
590 15:28:34
591 15:28:34
592 15:28:34
593 15:28:34
594 15:28:34
595 15:28:34
596 15:28:34
597 15:28:34
598 15:28:34
599 15:28:34
600 15:28:34
601 15:28:34
602 15:28:34
603 15:28:34
604 15:28:34
605 15:28:34
606 15:28:34
607 15:28:34
608 15:28:34
609 15:28:34
610 15:28:34
611 15:28:34
612 15:28:34
613 15:28:34
614 15:28:34
615 15:28:34
616 15:28:34
617 15:28:34
618 15:28:34
619 15:28:34
620 15:28:34
621 15:28:34
622 15:28:34
623 15:28:34
624 15:28:34
625 15:28:34
626 15:28:34
627 15:28:34
628 15:28:34
629 15:28:34
630 15:28:34
631 15:28:34
632 15:28:34
633 15:28:34
634 15:28:34
635 15:28:34
636 15:28:34
637 15:28:34
638 15:28:34
639 15:28:34
640 15:28:34
641 15:28:34
642 15:28:34
643 15:28:34
644 15:28:34
645 15:28:34
646 15:28:34
647 15:28:34
648 15:28:34
649 15:28:34
650 15:28:34
651 15:28:34
652 15:28:34
653 15:28:34
654 15:28:34
655 15:28:34
656 15:28:34
657 15:28:34
658 15:28:34
659 15:28:34
660 15:28:34
661 15:28:34
662 15:28:34
663 15:28:34
664 15:28:34
665 15:28:34
666 15:28:34
667 15:28:34
668 15:28:34
669 15:28:34
670 15:28:34
671 15:28:34
672 15:28:34
673 15:28:34
674 15:28:34
675 15:28:34
676 15:28:34
677 15:28:34
678 15:28:34
679 15:28:34
680 15:28:34
681 15:28:34
682 15:28:34
683 15:28:34
684 15:28:34
685 15:28:34
686 15:28:34
687 15:28:34
688 15:28:34
689 15:28:34
690 15:28:34
691 15:28:34
692 15:28:34
693 15:28:34
694 15:28:34
695 15:28:34
696 15:28:34
697 15:28:34
698 15:28:34
699 15:28:34
700 15:28:34
701 15:28:34
702 15:28:34
703 15:28:34
704 15:28:34
705 15:28:34
706 15:28:34
707 15:28:34
708 15:28:34
709 15:28:34
710 15:28:34
711 15:28:34
712 15:28:34
713 15:28:34
714 15:28:34
715 15:28:34
716 15:28:34
717 15:28:34
718 15:28:34
719 15:28:34
720 15:28:34
721 15:28:34
722 15:28:34
723 15:28:34
724 15:28:34
725 15:28:34
726 15:28:34
727 15:28:34
728 15:28:34
729 15:28:34
730 15:28:34
731 15:28:34
732 15:28:34
733 15:28:34
734 15:28:34
735 15:28:34
736 15:28:34
737 15:28:34
738 15:28:34
739 15:28:34
740 15:28:34
741 15:28:34
742 15:28:34
743 15:28:34
744 15:28:34
745 15:28:34
746 15:28:34
747 15:28:34
748 15:28:34
749 15:28:34
750 15:28:34
751 15:28:34
752 15:28:34
753 15:28:34
754 15:28:34
755 15:28:34
756 15:28:34
757 15:28:34
758 15:28:34
759 15:28:34
760 15:28:34
761 15:28:34
762 15:28:34
763 15:28:34
764 15:28:34
765 15:28:34
766 15:28:34
767 15:28:34
768 15:28:34
769 15:28:34
770 15:28:34
771 15:28:34
772 15:28:34
773 15:28:34
774 15:28:34
775 15:28:34
776 15:28:34
777 15:28:34
778 15:28:34
779 15:28:34
780 15:28:34
781 15:28:34
782 15:28:34
783 15:28:34
784 15:28:34
785 15:28:34
786 15:28:34
787 15:28:34
788 15:28:34
789 15:28:34
790 15:28:34
791 15:28:34
792 15:28:34
793 15:28:34
794 15:28:34
795 15:28:34
796 15:28:34
797 15:28:34
798 15:28:34
799 15:28:34
800 15:28:34
801 15:28:34
802 15:28:34
803 15:28:34
804 15:28:34
805 15:28:34
806 15:28:34
807 15:28:34
808 15:28:34
809 15:28:34
810 15:28:34
811 15:28:34
812 15:28:34
813 15:28:34
814 15:28:34
815 15:28:34
816 15:28:34
817 15:28:34
818 15:28:34
819 15:28:34
820 15:28:34
821 15:28:34
822 15:28:34
823 15:28:34
824 15:28:34
825 15:28:34
826 15:28:34
827 15:28:34
828 15:28:34
829 15:28:34
830 15:28:34
831 15:28:34
832 15:28:34
833 15:28:34
834 15:28:34
835 15:28:34
836 15:28:34
837 15:28:34
838 15:28:34
839 15:28:34
840 15:28:34
841 15:28:34
842 15:28:34
843 15:28:34
844 15:28:34
845 15:28:34
846 15:28:34
847 15:28:34
848 15:2
```

Task 3: Polish the CI/CD

I update the CD stages to be able to deploy the web application using Ansible, updated the pipeline to support multi-branch (e.g. master and develop) and jobs should be triggered based on the specific target branch.

Also for all of these I modified my configs for new keywords such as `cache`, `artifact`, `needs`, and `dependencies` to have more control of pipeline execution.



In github page for the project, you can observe all files containing version of completed 3 tasks.

I created folder ansible in the root of spa: `inventory.ini`, `deploy-playbook.yml`. Installed ansible on `VM2` and tested it running.

Changed `.gitlab-ci.yml` configuration for ansible (part of changes). Updated cache, artifact, needs, and dependencies to have more control of pipeline execution

```

GNU nano 7.2                               .gitlab-ci.yml *
DOCKER_IMAGE_NAME: "nikitanekhay/st19-spa"
DOCKER_IMAGE_TAG: "${CI_COMMIT_SHORT_SHA}"
ANSIBLE_FORCE_COLOR: "true"

# TO REUSE DEPENDENCIES ACROSS JOBS
cache:
  key: "${CI_COMMIT_REF_SLUG}"
  policy: pull-push

# CI Stage 1: Build the application
build-app:
  stage: build
  tags:
    - st19-runner
  cache:
    key: "${CI_COMMIT_REF_SLUG}"
    policy: pull-push

```

Here is push made from new branch, successfully deployed as it seen on the screenshots:

```

To st19.sne.com:root/st19-repo.git
* [new branch]      dev2 -> dev2
root@UbuntuDesktop:/home/nikita/Documents/cicdlab2/simple_pwa#

```

fix ansible ssh error
 ✓ Passed Created 1 day ago by Administrator, finished 1 day ago
 For commit 8b97cfc69

In **main**
 latest (branch) eo 6 jobs ⌚ 58 seconds, queued for 6 seconds

Pipeline Jobs 6 Tests 0

Group jobs by Stage Job dependencies

build	test	docker-build	docker-push	deploy
✓ build-app	✓ test-app	✓ build-docker	✓ push-docker	✓ deploy-production ✓ rollback-production

